What Communication Does to a Team of Agents

A characterization of multi-agent round dynamics, with emphasis on correct minorities

ARD36 study ▪ 5 agents ▪ 7 protocols ▪ 4 rounds ▪ Qwen3-8B/32B ▪ MATH-500 + MMLU-Pro

Full report over the repaired tables (`tables_v2`). Generated September 13, 2026.

The result, in one paragraph

A multi-agent team that lets its members read each other does *not* find more right answers than a team whose members only revise privately. It finds *fewer*. Peer visibility costs **4.6 percentage points** of team truth-coverage on a shared-root, item-paired design ($p = 0.008$, Holm 0.024, **rejects** as a pre-registered primary). The mechanism is **consolidation, not discovery**: communication concentrates agreement inside whichever answer the team already holds — the *mean* number of correct agents rises while the probability that *anyone* is correct falls. The same signature appears independently in the residual stream, where peer exchange contracts the team's representational spread, and the contraction deepens monotonically with network depth.

Read this before any number below

This study was originally written up as **"all three pre-registered contrasts are null."** That was wrong. A cell-keying defect caused a 16-draw sampling bank to overwrite the round-0 baseline of the PRIVATE arm, and the defect had a direction. After repair, one primary inverts its sign and rejects. **Every number in this report comes from the repaired tables**; the frozen originals are quoted only to show what moved.

1 What this document is

Between generation and analysis, this study accumulated a measurement problem large enough to invert one of its own registered results. This report is the output of repairing that problem and then re-characterizing the data from scratch.

Provenance. Every figure and number derives from `tables_v2/`, rebuilt from the raw cell outputs after ten repairs (§2). The frozen `tables/` are byte-identical to their state before this work began — verified by SHA-256 on all 19 artifacts. The regression suite is **1,519 passed / 1 skipped / 17 subtests**, up from a 1,336-test baseline, with zero regressions.

The evidence base. 67,104 agent decisions across 4,888 episodes; 51,360 graded answers; 240 held-out test items per arm; seven communication protocols sharing a byte-identical round-0 root bank. Analysis enumerated **1,441 candidate trends**, of which **692 were estimable**; 56 were computed and 39 were put through adversarial verification.

How claims were vetted. Each computed claim was handed to an independent verifier instructed to *refute* it, attacking composition first. The yield was **35 weakened, 4 refuted, 0**

confirmed as originally stated. Wherever a verifier disagreed with an analyst, the verifier's wording is what appears here.

2 The repair, and why it mattered

2.1 A defect that inverted a registered result

`analysis.coverage_change_arm` accumulated per-episode coverage into a dictionary keyed on `episode_id` alone, and overwrote on collision. A ROOTS cell — the 16-draw independent sampling bank — defaults its protocol to PRIVATE and shares an `episode_id` with the real PRIVATE episode of the same item. The bank's round-0 coverage therefore replaced the five-agent team's round-0 coverage *in the PRIVATE arm only*, inflating its baseline until PRIVATE appeared to *lose* coverage.

Table 1: The three pre-registered primary contrasts, before and after repair.

Contrast	v1 (frozen)	v2 (repaired)
GATE − PEER (r3 accuracy)	−0.0083, p=0.465	−0.0089, p=0.396
GATE − RANDOM_PROTECT	−0.0258, p=0.056	−0.0226, p=0.035, Holm 0.071
peer − private (coverage)	+0.0125, p=0.535	**−0.0458, p=0.008**, Holm **0.024, rejects**

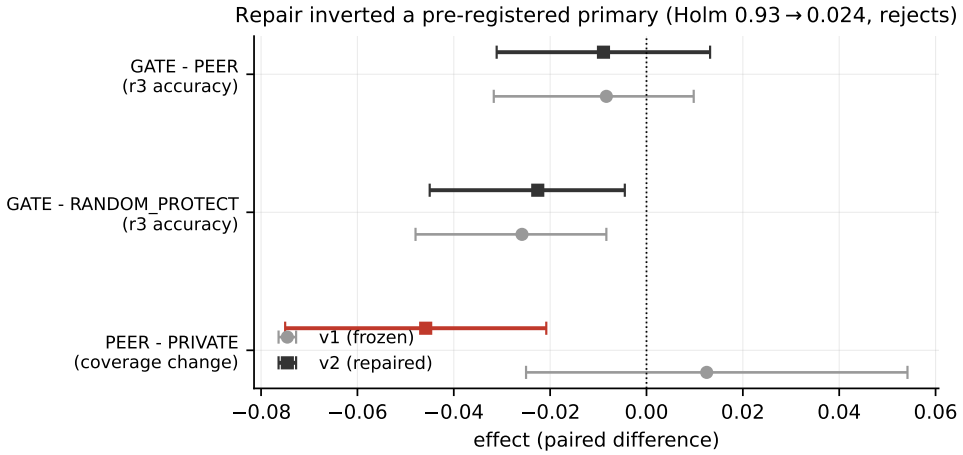


Figure 1: The repair moved one contrast across zero. The isolation is clean: PEER's change is *byte-identical* in v1 and v2 (−0.02083), because no ROOTS cell aliases a PEER episode. Only PRIVATE moved, from −0.0333 to +0.0250.

Three independent routes to the same number

The repaired estimator gives −0.04583. A direct trajectory computation gives (77.92−80.00)−(82.50 − 80.00) = −4.58 pp. A root-paired exact McNemar on identical round-0 roots gives **12 vs 1 discordant,** $p = 0.0034$. Three different code paths, one answer.

2.2 The other nine repairs

Table 2: Repairs applied, each with the check that would have caught it failing.

Repair	Verification
Cell-kind keying (M2)	`(cell_id, episode_id, round)` unique on 12,832/12,832
Tri-state correctness (M1)	5,047 ungradeable no longer scored wrong; $C_{t+1} - C_t = WC - CW$ holds 6,771, **violated 0**
GPU-second attribution (B-1)	**268.75** request-hours → **25.41** attributed GPU-hours recovered
Fork re-metering	1,440/1,440 recovered; FLOPs byte-identical; 0 mismatches on 263 control rows
Transition change rule	**40.0%** of `changed_class` was validity flips, not switches
pass@k convention	complete-case + Manski bounds on a fixed 171-item set
Minority frame	PRIVATE round-0 rescued: **320** → **2,720** rows
Receiver assay	"unexecuted" → **all 1,280 probes graded**
Calibration	real item clustering restored (v1 had *one* pseudo-cluster)
Selectors	diagnosed: three distinct causes, none a wiring bug

Two repairs that removed findings rather than adding them

Calibration. v1 reported a within-item AUC of 0.7555 for self-reported confidence. But v1's `decisions` table had no `item_id` column at all, so every row's item was `None` and all rows collapsed into one pseudo-cluster. The real figure is **0.5723**. Verbal confidence barely ranks correct above incorrect *within* an item; the old number was between-item difficulty. All three *system* signals' within-item AUC is now **undefined** — a team row is one candidate per item, so a mixed pair cannot exist at that grain.

Selectors. Three of five selectors emit `selected_correct` entirely null (120,024 of 200,040 rows). This is what manufactures "0 out of N" artifacts. Causes differ: `llm_audit` needs generation that never ran (0 of 934 cell directories); `CURRENT` and `dynamics_only` need a model head that was never fitted. Running every cell the spec names would still leave two of the three null.

3 The central result

3.1 Communication destroys truth-coverage

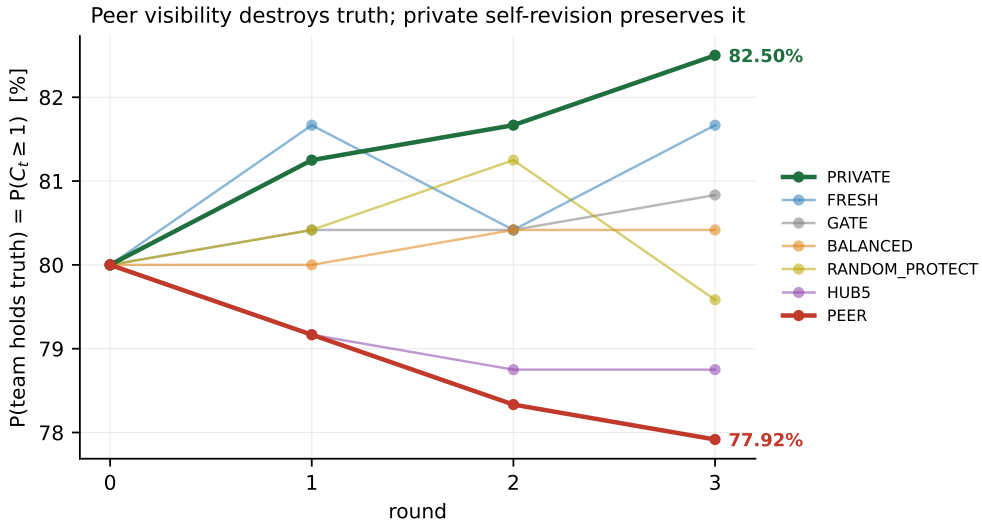


Figure 2: P(at least one of five agents holds the correct answer), by round. All seven arms start at exactly 80.00% because they share a byte-identical round-0 root bank — an exact paired design. PEER falls to 77.92%; PRIVATE rises to 82.50%. $n = 240$ items, 240 episodes per arm.

3.2 The mechanism: consolidation, not discovery

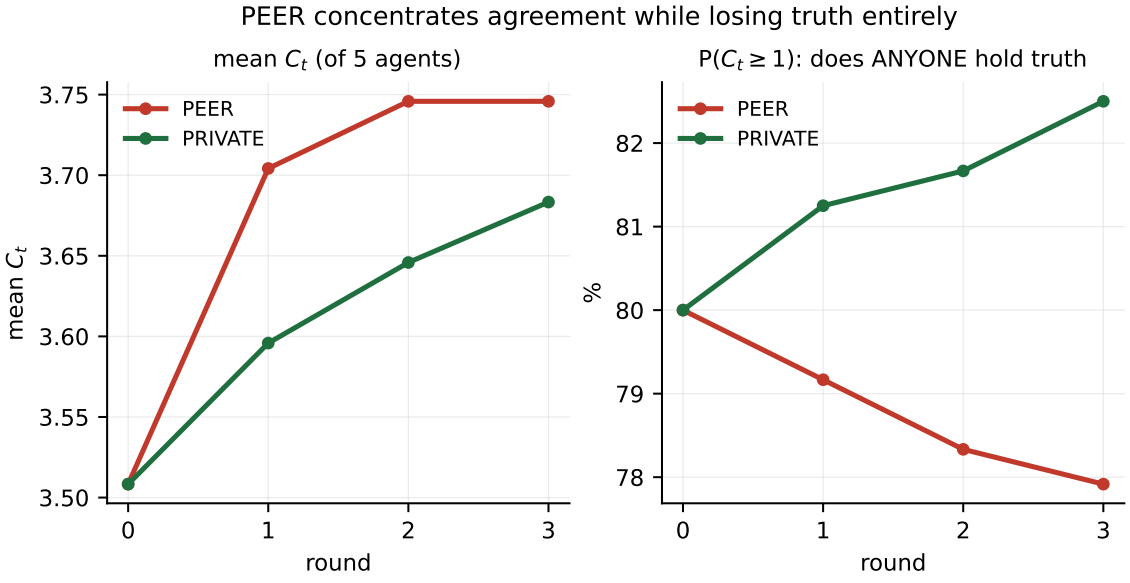


Figure 3: The dissociation that names the mechanism. Under PEER the *mean* number of correct agents **rises** (3.508 → 3.746) while the probability that *anyone* is correct **falls** (80.00 → 77.92%). Communication amplifies whatever the plurality already holds.

Both absorbing corners at once

The top-ranked finding of the characterization generalizes this beyond the registered contrast. Wrong-unanimity prevalence *rises* +5.40 pp [+3.47, +7.61] in the five packet-passing arms and is flat in both zero-packet arms (PRIVATE −0.74, FRESH −0.26, both null). The root-paired difference-in-differences on covered items gives COMM − PRIVATE = −4.23 pp [-6.24, -2.46] — independently reproducing the primary's magnitude from a *different outcome* on a *different panel* (384 items).

3.3 Consensus is what locks in the error

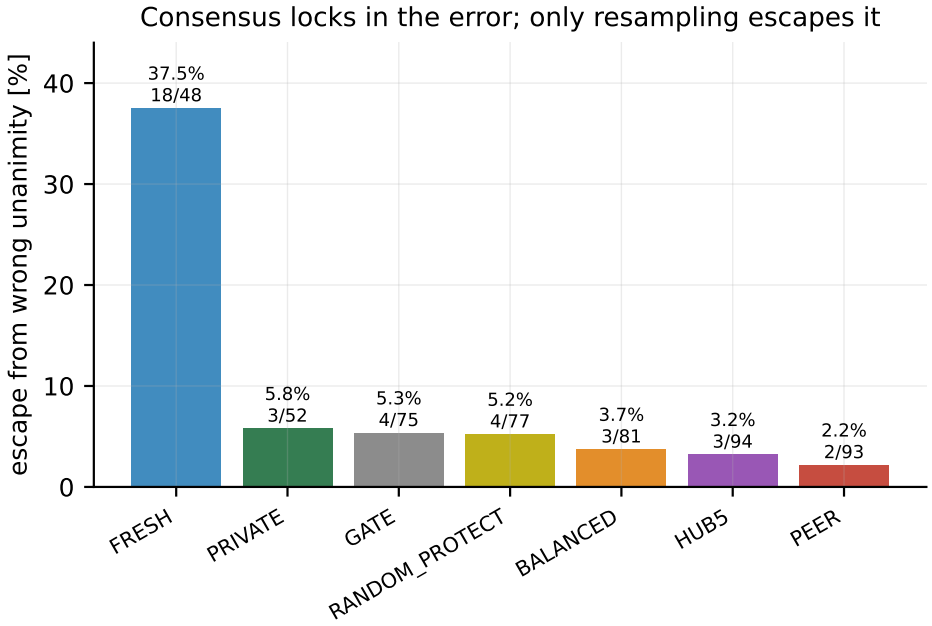


Figure 4: Escape rate from wrong unanimity. Under every communicating protocol a team that agrees on a wrong answer essentially never recovers (2.2–5.8%). Under FRESH — independent resampling with no message passing — it escapes **37.5%** of the time.

The comparison localizes the failure precisely. It is not that the model cannot find the answer: resampling the same model on the same items recovers the truth an order of magnitude more often. It is that *consensus* is absorbing.

A clean structural fact. Across **5,861 transitions out of any unanimous state** in the entire study — correct or wrong, communicating or not — exactly **zero** land on a different unanimous class. Once a team agrees, it never coordinates onto a *new* answer. It can only decay.

4 Communication versus sampling

This is the comparison practitioners actually face: is a team of k agents worth more than k independent samples?

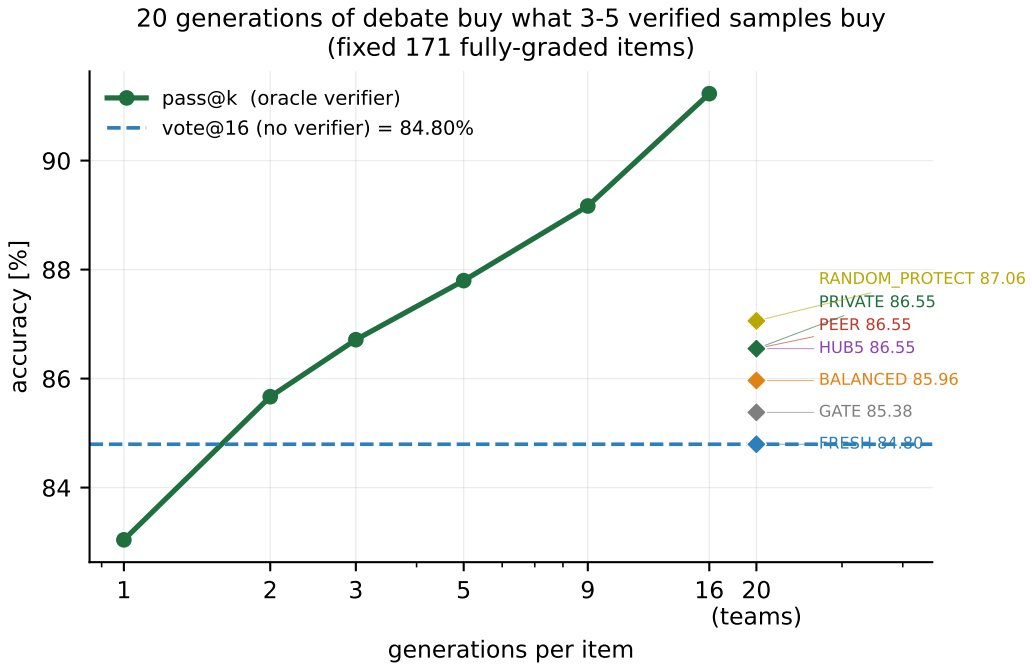


Figure 5: Accuracy against generations spent, on the fixed 171-item set where all 16 bank draws are graded (so both sides use one convention). Every multi-agent arm lands between pass@2 and pass@5 while spending **20 generations**. FRESH at round 3 equals vote@16 *exactly* (84.80%).

What debate is worth

Against an **oracle verifier**, a 5-agent team running 3 revision rounds (20 generations) reaches what **3–5 verified samples** reach — a 4–7× compute disadvantage. Against the achievable comparator **vote@16** (84.80%, no verifier), the best arm gains ≈ 2.3 pp. And FRESH — resampling without exchange — lands exactly on vote@16, which says the gain comes from *revision*, not from more samples.

A measurement trap that nearly produced a false headline

After the tri-state repair, team accuracy stopped counting ungradeable answers as wrong — but `bank_summary`'s pass@k still did, because `bool(None)` is `False`. The two halves of this comparison were briefly on *different conventions*, biasing pass@k downward against the teams. Uncorrected, it would have read as "revision beats best-of-5." A second trap sits underneath: complete-case pass@k over all 240 items silently shrinks the item set as k grows (236 items at k=1, only **171** at k=16), because a bank needs k graded draws. Every number above uses the fixed 171-item set.

5 What "revision" actually consists of

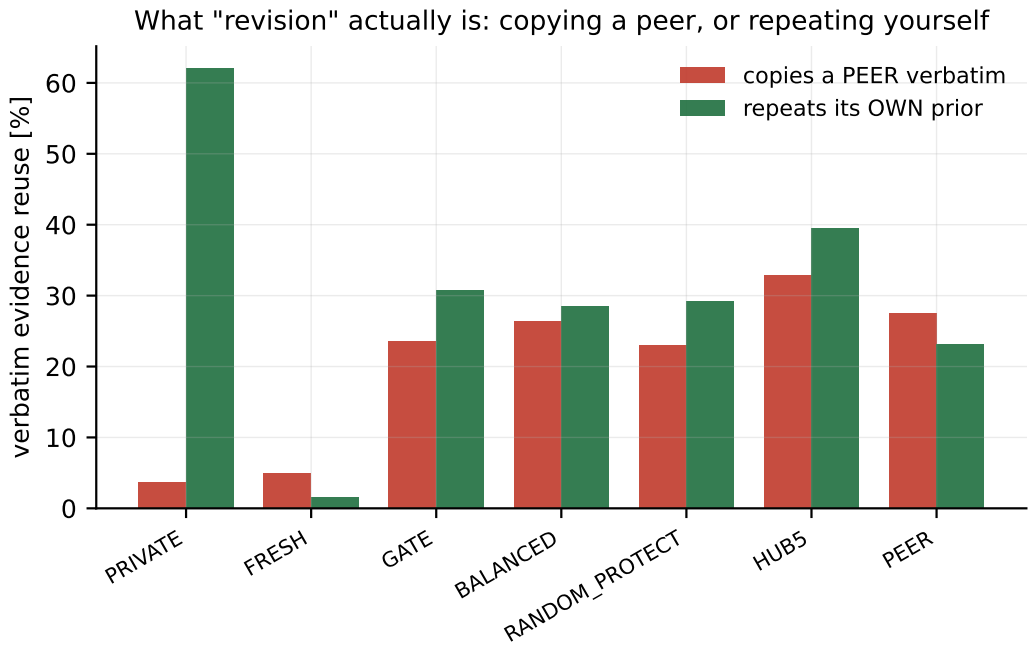


Figure 6: Verbatim reuse of evidence text between consecutive rounds. Communicating arms import a peer's evidence byte-identically 23–33% of the time against a 4.7–4.9% floor; under PRIVATE, agents re-emit *their own* prior evidence **62%** of the time.

Peer copying is caused by the channel, not by item-level text determinism: a placebo that matches against a *different episode's* agent on the same item gives 1.1%, against PEER's 20.4% in-episode rate. And PRIVATE's advantage is not thoughtful reconsideration — among agents already holding a correct answer it repeats itself verbatim 70.2% of the time. The truth is preserved by *inertia*.

A mechanism that failed

The natural hypothesis — that correct minorities resist copying and that is why they survive — is **not supported**. Correct minorities do not copy less than the wrong bloc; the point estimate has the opposite sign and the interval covers zero (peer copy +0.0367 [-0.0361, +0.1097], MDE 0.0677, 100 paired units). Reported because it was predicted and failed.

6 The neural correlate

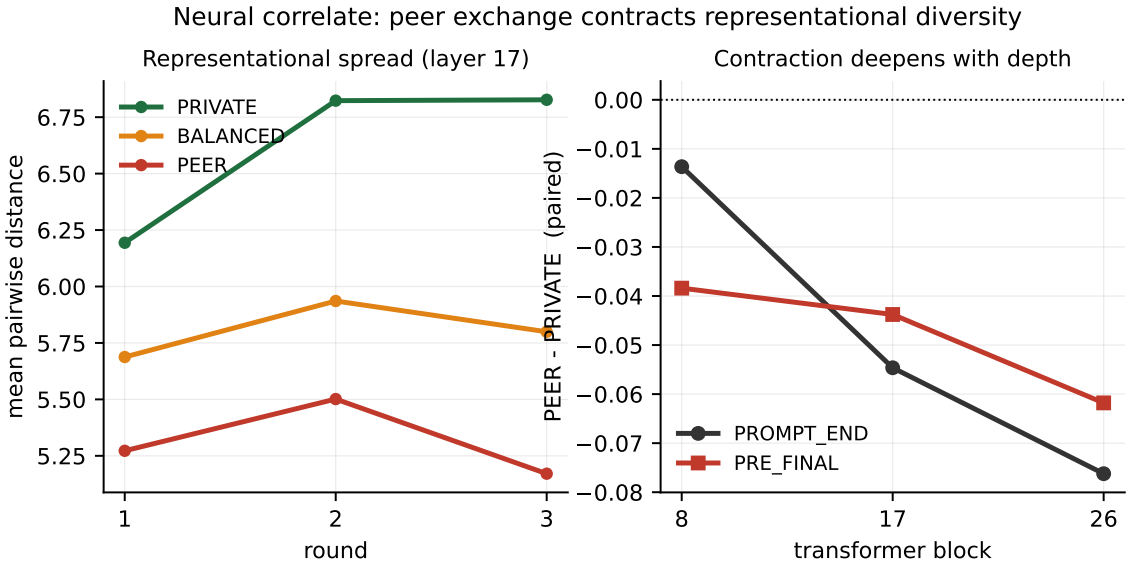


Figure 7: Residual-stream geometry from teacher-forced replay of 7,200 stored decisions (no regeneration). **Left:** representational spread across the five agents by round — PEER contracts, PRIVATE expands. **Right:** the paired PEER − PRIVATE contraction, by transformer block, at both prospective anchors.

Within-item paired PEER − PRIVATE at round 3, item-clustered bootstrap, $n = 144$ items (layer 17, PRE_FINAL):

Metric	Effect	95% CI
mean pairwise distance	-0.0438	$[-0.0555, -0.0334]$
max pairwise distance	-0.0668	$[-0.0862, -0.0490]$
leading eigenvalue share	-0.0548	$[-0.0797, -0.0300]$
mean displacement alignment	$+0.0539$	$[+0.0311, +0.0769]$

The effect holds at **all six** layer × anchor combinations, and at PROMPT_END it deepens monotonically with depth ($-0.0136 \rightarrow -0.0546 \rightarrow -0.0762$ across blocks 8/17/26) — which rules out a layer-specific artifact. BALANCED, which hides how many agents hold each answer, sits between the two ($+0.0169\,[+0.0093, +0.0252]$ against PEER), in the same direction as its behavioural effect.

What this is not

Descriptive only. No probe, no prediction, no causal claim. A probe separating a correct minority from its wrong bloc would have $\mathbf{MDE_{80}(AUC)} \approx 0.73$ on this panel — neither a hit nor a null would be informative — so it was not attempted. The geometry and the behaviour are not independent *evidence*: they describe one process on one set of episodes. Treat it as mechanism illustration.

7 The correct minority

Detectable in principle, useless in practice

Inside a contested, fully-graded team, the correct minority is separated from the wrong bloc by `answer_logprob`: +0.0220 [+0.0091, +0.0350] within-team-round paired — described by its verifier as "the most robust thing in the correct-minority panel." Self-reported confidence gives +0.0115 [+0.0007, +0.0223], a fifth of one grid atom. Both are real. Both are far too small to act on. A flat "indistinguishable" null would not survive review, and would be false.

Two qualifications the verifier attached, and they matter. The logprob gap *halves* to +0.0115 [+0.0006, +0.0224] once `final_tokens` is controlled — part of it is answer length, not conviction. And a placebo that forces the correct bloc to be the *majority* finds the gap is *larger*, not absent (n_{correct}=3: +0.0317 [+0.0102, +0.0533]). So this is a signature of **correctness in general, not of minority-ness**. It cannot be used to find embattled correct agents specifically.

Two quantities that are routinely conflated, and must not be:

- A lone correct agent **converts at least one teammate in 37.7%** of rounds (101/268).
- The team's plurality is correct in only **22.3%** of lone-correct rounds (65/291).

Persuasion is common. Winning the vote is not.

8 Where correctness is decided

The 16-draw bank is generated before any communication, so bank statistics are *exogenous* to the panel — the instrument that breaks the circularity of using in-panel accuracy as a difficulty proxy. Scoring rounds 1–3 only, so bank draws never score themselves:

Minority-correctness concentrates where the bank *contains* the truth but its own plurality vote *discards* it. At bank `selection_gap`=1: $P = 6.10\%$ [3.38, 9.02] versus 0.47% [0.21, 0.79] at gap=0 — a **13× rate ratio**, identified without touching the panel.

9 The receiver assay

1,280 single-agent probes, previously recorded as *unexecuted*, graded here for the first time with an independent mod-97 evaluator: **947 correct / 333 incorrect**.

Table 3: Receiver arms. The correct answer is endorsed 4–1 (MAJORITY) or 1–4 (MINORITY); CHECKED arms claim verification. $n = 256$ per arm.

Arm	Accuracy	@1024 tokens	@4096 tokens
PRIVATE_TRACE (control)	0.668	0.500	0.836
MINORITY_UNCHECKED	0.738	0.563	0.914
MINORITY_CHECKED	0.746	0.578	0.914
MAJORITY_CHECKED	0.766	0.586	0.945
MAJORITY_UNCHECKED	0.781	0.586	0.977

Budget dwarfs social framing

On the control arm alone, quadrupling the thinking budget moves accuracy from **0.500 to 0.836**. Against that, the entire social manipulation — whether the correct answer is endorsed by four peers or one — is worth +5.86 pp, and a claim of verification is worth nothing measurable (−0.39 pp [-4.30, +3.52]). The single largest lever on a reasoning agent in this study is how long it is allowed to think.

10 Methodological findings

The pairing is doing all the work

Item ICC is **0.19–0.84** (design effect ×21 to ×147). A marginal two-arm protocol contrast has an MDE of **0.17–0.29**. The same contrast paired within item has an MDE of **0.021–0.029** — a 6–8× gain, and the only reason a −0.046 effect is detectable at all. Any multi-agent study that compares arms without pairing on the item is underpowered by roughly an order of magnitude and will not know it.

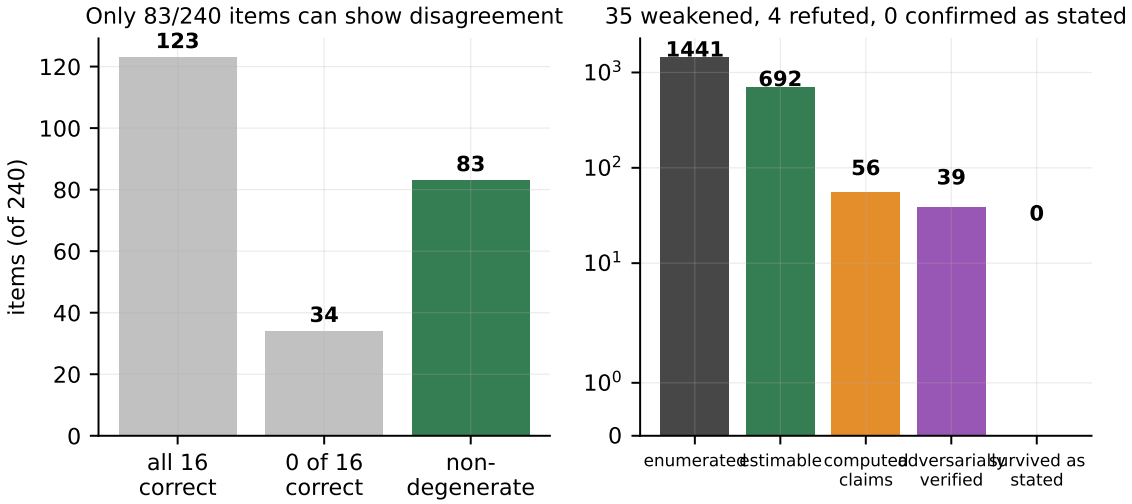


Figure 8: Left: the panel's usable range — 123 of 240 items are solved by all 16 draws and 34 by none, leaving **83** that can express disagreement at all. **Right:** the verification funnel, log scale.

Composition traps found in this data. Ungradeable answers coerced to "wrong" nearly *double* the measured truth-loss hazard (13.1% vs 23.7% at C_{t-1}=1). "Minority correctness decays over rounds" is at-risk pool migration, not a trend. A "switch rate" read off `changed_class` overstates switching by two thirds. Three of five selectors are structurally empty, so any "0 of N" computed on them is an artifact of an absent column.

The panel affords exactly one contrast. Of 21 protocol pairs, only 12 clear the minimum-events floor, and 7 of those are null with MDEs of 2.9–5.6 pp. The significant result is not selected

from many — there was only ever room for one.

11 Coverage: what was and was not computed

Table 4: The frozen enumeration manifest. Coverage is arithmetic, not a claim.

Bucket	Cells	Share
Enumerated (131 measurables × 11 axis roles)	1,441	100.0%
Estimable	692	48.0%
Structurally void (axis cannot vary in that slice)	463	32.1%
Known-degenerate (censored or determined measures)	143	9.9%
Not estimable (below the events/items floor)	55	3.8%
Redundant	55	3.8%
Identity (algebraically forced)	33	2.3%

Cells pruned as *identity* are worth naming, because each would otherwise look like a finding: $P(\text{agent correct} \mid n_{\text{correct}}{=}k) = k/5$ exactly; all four minority senses are subsets of `correct`=1; protocol differences at round 0 are identically zero, measured on all 310 shared items.

12 Limits

1. **One model family, one scale for the main result.** Qwen3-8B carries every headline; 32B ran three protocols on 96 items. Minority-correctness at 32B is **15 items**, below the floor — so nothing minority-conditioned can be said about scale.

2. **The headline rests on 13 discordant items**, exactly at the minimum for a paired exact test. It reproduces three ways, but it is one contrast on one family.

3. **Identification is from randomized protocol assignment, not from the baseline.** Because round 0 is bit-identical across arms, the pre-registered "root→r3 change" is algebraically the r3 *level* contrast. The shared root buys variance reduction, not identification. A reader who takes it for a difference-in-differences has been misled.

4. **35 of 39 verified claims were weakened.** That is the honest yield of adversarial verification, and it should temper confidence in the 17 computed claims that were never verified.

5. **Two things were not run.** The challenge cohort (E2C), which would test the mechanism interventionally by constructing teams at known initial correct-counts, has only 27 eligible items under the repaired grading rule — below the testability floor. Spectral entropy came back null in the geometry tables.

13 What would falsify this

The consolidation claim predicts that any manipulation *reducing* a team's exposure to peer content should preserve coverage, monotonically in dose. Three tests are available and were not run: E2C at constructed initial correct-counts $C_0 \in \{1, 4\}$; the randomized `protected` slot extended

beyond RANDOM_PROTECT; and a capture of the protected arm, which would let the geometry contraction be measured under randomized exposure rather than observationally. If coverage does *not* track exposure dose under those, the mechanism as stated is wrong.

Reproduce. `PYTHONPATH=src python -m agents_scaling.ard.compact --run-root $R --out-dir tables_v2`, then `.minority`, `.analysis --tables-dir tables_v2`, `.keys`, `.panels --check`. Figures: `report/figs/`. Full trend table: `analysis_v2/TRENDS.md` (2,295 lines). Repair register: `docs/ard36/05_stage0_repair.md`.